

Grounded but Brittle: UMLS-Based Verification and the Concept Mismatch Problem in Neuro-Symbolic Medical Question Answering

Working draft — Paper 2 (MedQA2026). [TBD-EXP#] items await neuro-symbolic experiments; LLM-only baseline is verified.

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Title (working; alternates below)

Grounded but Brittle: UMLS-Based Verification and the Concept Mismatch Problem in Neuro-Symbolic Medical Question Answering

Alternates: 1. *The Concept Mismatch Problem: When Symbolic Biomedical Knowledge Fails to Ground LLM Medical Reasoning.* (concept-mismatch focus) 2. *Hybrid Neuro-Symbolic Reasoning for MedQA: Combining UMLS Grounding with Language-Model Flexibility.* (neuro-symbolic focus) 3. *Grounding LLM Medical Reasoning in UMLS: How Far Does Symbolic Knowledge Get Us?* (UMLS-grounding focus) 4. *Auditing Medical Answers: UMLS-Grounded Chain-of-Verification for Detecting Wrong*

LLM Reasoning. (verification focus) 5. Structured Knowledge Meets Language Models: A Controlled Study of UMLS Grounding and Verification for Medical Question Answering. (biomedical-informatics venue)

Abstract

Large language models answer USMLE-style medical questions well, but their reasoning is hard to audit: they produce fluent rationales with no guarantee that the underlying clinical concepts and relations are correct. Structured biomedical knowledge — notably the Unified Medical Language System (UMLS) — offers grounding, traceability, and contradiction detection, yet exact symbolic matching is brittle when clinically equivalent ideas are phrased differently, the *concept mismatch problem*. We present a controlled study, on a category-balanced set of 1,030 MedQA-style questions (206 each across Clinical Findings, Diagnosis, Mechanism/Pathophysiology, Next Step/Workup, Treatment/Management), of how UMLS grounding and symbolic verification help, fail, and combine with LLM reasoning. We compare LLM-only, symbolic-only (concept-overlap and relation-aware), UMLS-in-prompt, verifier reranking, UMLS-grounded Chain-of-Verification (with and without revision), and a hybrid decision module that fuses LLM likelihood with symbolic support and violation penalties. We quantify grounding coverage, contradiction and semantic-type violations, a concept-mismatch taxonomy, and whether symbolic scores predict LLM errors. We expect, and motivate from existing data, that pure symbolic scoring is far below LLM-only accuracy [TBD-EXP3], that concept mismatch is a dominant symbolic failure mode (canonical concepts appear under six or more surface forms in this benchmark), that verification improves weak proposers while naive answer revision can harm [TBD-EXP6], and that grounding and violation scores predict LLM correctness above chance [TBD-EXP9]. The contribution is an empirical characterization of when and why symbolic biomedical knowledge helps versus fails, and a hybrid framework that uses UMLS to *audit and constrain* rather than replace LLM reasoning. We release data, configurations, and analysis code. We make no claims of clinical utility.

1. Introduction

P1 — Accuracy alone is insufficient. On MedQA, strong base LLMs already score well above chance (Paper 1: up to 73.4% at greedy decoding on this balanced set), but a correct letter says nothing about whether the model reasoned over the right clinical concepts. For medical reasoning, *how* an answer is reached — and whether it can be audited — matters as much as the answer.

P2 — Why grounding and verification matter. Auditable medical reasoning requires linking claims to defined clinical concepts and checking them for support, contradiction, and plausibility. This is precisely what free-form LLM rationales do not guarantee.

P3 — Why UMLS is attractive. UMLS provides millions of concepts (CUIs), semantic types, definitions, and curated relations (e.g., `may_treat`, `manifestation_of`, `contraindicated_with`). In principle it offers exactly the grounding, traceability, and contradiction-detection that LLMs lack.

P4 — Why pure symbolic UMLS reasoning is brittle. In practice, exact concept matching is fragile. Our seed analysis shows that the amount of groundable surface in a question is *uncorrelated* with difficulty (per-question $r = -0.02$), so naive concept overlap cannot explain — let alone fix — where models fail. Preliminary repository experiments put pure symbolic accuracy far below LLM-only [TBD-EXP3].

P5 — The concept mismatch problem. A symbolic system may fail to connect “chest discomfort” with “chest pain,” or “shortness of breath” with “dyspnea.” We quantify this: in this benchmark, dyspnea appears under six distinct surface forms across 113 questions, of which the canonical term “dyspnea” covers only 28 — so exact matching to one surface would miss roughly three-quarters of clinically equivalent mentions.

P6 — Proposed hybrid framework. We propose a pipeline (Figure 1) in which UMLS grounds and verifies while an LLM bridges semantic gaps: question parsing, UMLS grounding, symbolic candidate scoring, LLM rationale generation, UMLS-grounded Chain-of-Verification, and a hybrid decision module emitting an answer, a confidence proxy, and an evidence trace.

P7 — Summary of experiments. We compare LLM-only, symbolic-only, UMLS-in-prompt, verifier reranking, CoVe (verify-only and with revision), and the hybrid module across the five reasoning categories, with grounding/verification metrics, a concept-mismatch taxonomy, ablations, and a test of whether symbolic scores predict LLM errors.

P8 — Contributions (Section 1.1).

1.1 Contributions

1. A **controlled comparison** of LLM-only, symbolic-only, and hybrid neuro-symbolic systems on a category-balanced MedQA set, with proper statistics.
2. An **empirical characterization of the concept mismatch problem**, including a labeled taxonomy and quantified surface-form prevalence in the benchmark.
3. A **UMLS-grounded verification analysis** establishing where symbolic verification helps a reasoning proposer and where answer revision harms it.
4. Evidence on whether **grounding coverage and symbolic violation scores predict LLM errors**, connecting symbolic signal to uncertainty estimation.
5. A released, reproducible **hybrid framework** that uses UMLS to audit and constrain — not replace — LLM medical reasoning. We make no clinical-deployment claims.

2. Related Work

Clusters, what to cite, positioning, and the gap.

Medical QA benchmarks (MedQA/USMLE — Jin et al. 2021; MedMCQA; PubMedQA). *Gap:* scored in aggregate, no grounding/verification analysis. **LLMs for clinical reasoning** (Med-PaLM, GPT-4 medical evaluations). *Position:* we audit reasoning rather than chase accuracy. **Biomedical knowledge graphs and UMLS/SNOMED CT/MeSH concept normalization** (Bodenreider 2004; MetaMap; entity-linking literature). *Position:* we use UMLS as a grounding/verification substrate and characterize its brittleness. **Neuro-symbolic AI** (knowledge-constrained neural reasoning). *Position:* a concrete medical instantiation. **Retrieval-augmented / knowledge-grounded medical QA.** *Position:* we contrast prompt-injection with constraint/verification. **Chain-of-Verification & self-verification** (Dhuliawala et al.). *Position:* we ground verification in a knowledge graph, not only self-checking. **Explanation faithfulness & hallucination detection, and clinical decision support / symbolic reasoning.** *Gap this paper fills:* a controlled, category-resolved account of when UMLS grounding and symbolic verification help versus fail for LLM medical QA, centered on the concept mismatch problem.

3. Background: UMLS and Neuro-Symbolic Medical Reasoning

Paper-ready exposition. **UMLS** integrates biomedical vocabularies into concepts, each with a **CUI**, one or more **semantic types** (e.g., Disease or Syndrome, Sign or Symptom, Pharmacologic Substance), and curated **relations** between concepts. To **ground** medical text, mentions are linked to CUIs (entity linking), enabling traceable reasoning: an answer can be tied to specific concepts and the relations among them, and checked for contradictions or semantic-type incompatibility. Grounding supports **interpretability** but is **not** clinical reasoning: a correct CUI map does not entail a correct inference. **Concept normalization is hard** because the same clinical idea is expressed in many surface forms, abbreviations, and granularities; **exact symbolic matching fails** when wording diverges from canonical terms (our seed data: ≥ 6 surface forms per common concept). **LLMs** can bridge these gaps via paraphrase-robust representations, but **unconstrained LLMs** still hallucinate and cannot be audited — motivating grounding and verification as a check on, not a replacement for, neural reasoning.

4. Dataset

The benchmark is the focused balanced MedQA subset: **1,030** questions, **206 per category** across the five reasoning categories, 4-option (A-D), random baseline 25% (shared with Paper 1; see Paper 1 §3 for construction and the category-label validation plan, [TBD: EXP7 there]). Each item carries pre-extracted clinical phrases (metamap_phrases, mean 34.9/question), which we use both as grounding candidates and to quantify surface-form prevalence. **Table 1** reports the category distribution. We additionally define a **hard-core evaluation set** — the 144 questions (14.0%) that all five base models miss at greedy decoding (Clinical Findings 41, Diagnosis 30, Next Step 30, Mechanism 22, Treatment 21) — as the focus for measuring verifier/hybrid gains.

5. Methods

The pipeline of Figure 1; modules A-G. Build steps are EXP1-EXP10.

A. Question parser. Extract stem, the four options, the reasoning category, and clinical elements (demographics, symptoms, signs, labs, medications, history, temporal cues).

B. UMLS concept grounding (EXP1-EXP2). Map clinical mentions in the stem and each option to CUIs with semantic types, aliases, and confidence; handle negation/uncertainty (present/absent/uncertain). Emit grounding coverage and per-option grounding rate.

C. Symbolic candidate scoring (EXP3). Score each option by relation support (category-appropriate UMLS relations), semantic-type compatibility, and penalties for unsupported or contradictory relations; concept-overlap and relation-aware variants.

D. LLM reasoning generation (EXP4). Produce a structured rationale with an evidence table and, per option, an explicit support/contradiction list.

E. Chain-of-Verification (EXP6). Verify rationale claims against UMLS with five symbolic verifiers — semantic-type incompatibility, demographic implausibility, low grounding coverage, polarity/negation conflict, internal contradiction — optionally triggering revision.

F. Hybrid decision module (EXP7). Fuse LLM answer likelihood with symbolic support and violation penalties; apply gated revision; output a final answer and confidence proxy.

G. Output / evidence trace. Final answer with supporting and contradictory concepts, grounding coverage, verification flags, and a human-readable explanation trace.

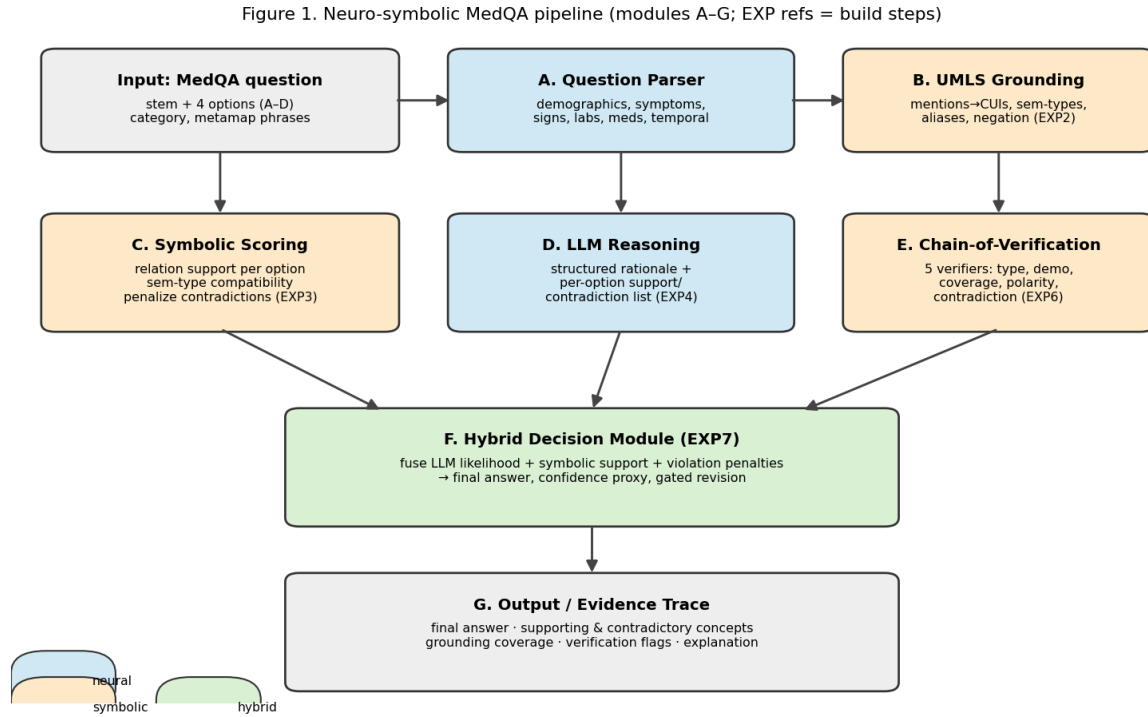


Figure 1. Neuro-symbolic MedQA pipeline (modules A-G). Build steps map to EXP1-EXP10.

6. Experimental Setup

Systems compared (Table 2). (i) random/majority baseline (25%); (ii) **LLM-only** (verified, Paper 1); (iii) pure symbolic — concept-overlap, relation-aware (EXP3); (iv) TF-IDF lexical baseline (EXP3); (v) UMLS-in-prompt (EXP4); (vi) UMLS verifier reranking (EXP5); (vii) CoVe verify-only and (viii) CoVe with revision (EXP6); (ix) hybrid decision (EXP7); and an oracle upper bound (best-of-systems) for context. **LLM components** use vLLM with the configs in configs/; **symbolic components** use the UMLS index from EXP1. **Statistics:** bootstrap CIs, McNemar for paired differences, paired bootstrap for category comparisons, logistic regression for correctness prediction, AUROC/AUPRC for error prediction, and category×method interaction tests — reusing the verified Paper 1 statistics harness. We will **avoid overclaiming**: small or mixed improvements will be reported with CIs and not framed as decisive.

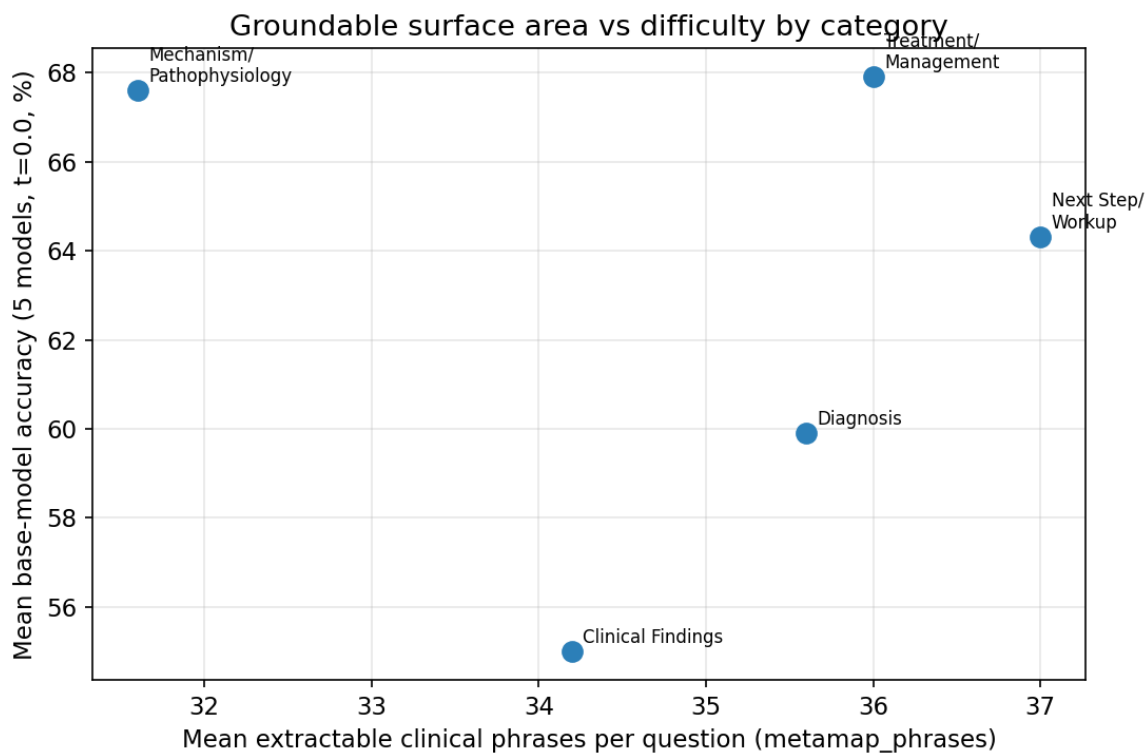
7. Results

All neuro-symbolic numbers are placeholders pending experiments; the LLM-only column is verified (Paper 1).

7.1 Overall accuracy by method (Table 3). LLM-only baseline (t=0.0): OLMo-3 7B

46.18 ... Qwen2.5 32B 73.40. Symbolic-only: [TBD-EXP3] (expected well below LLM-only; README anecdote ~30% pure UMLS, ~0% TF-IDF). UMLS-in-prompt [TBD-EXP4]; verifier rerank [TBD-EXP5]; CoVe verify-only / with-revision [TBD-EXP6]; **hybrid** [TBD-EXP7].

7.2 Category-specific accuracy (Table 4, Figure 4). [TBD] per-method $\times$ category; test whether symbolic grounding helps Diagnosis/Mechanism more than Treatment/Workup (BRIEF H7). Seed context: base difficulty ordering has Clinical Findings hardest, Mechanism/Treatment easiest, and difficulty is *uncorrelated* with groundable phrase count ($r = -0.02$).



Seed figure. Groundable surface area (mean extractable clinical phrases) vs base-model accuracy by category. Difficulty is not explained by how much there is to ground.

7.3 Grounding and verification metrics (Table 5). Grounding coverage and answer-choice grounding rate [TBD-EXP2]; verifier firing rates and violation distributions [TBD-EXP6]; verification precision/recall against manual labels [TBD-EXP6/EXP8].

7.4 Verification helps, revision can harm. [TBD-EXP6] — reproduce/qualify the README pattern (verify-only > baseline; revision $\leq$ verify-only) on the full balanced set with CIs.

7.5 Symbolic scores predict LLM errors (Figures 5, 6). [TBD-EXP9] AUROC for predicting LLM correctness from grounding coverage and violation scores; risk-coverage curve.

8. Concept Mismatch Analysis

A central section. Taxonomy in Table 6; examples in Figure 7; labels from EXP8.

We organize symbolic failures into a taxonomy: lexical-synonym, clinical-paraphrase, granularity, symptom $\leftrightarrow$ diagnosis, mechanism $\leftrightarrow$ disease, treatment-indication, lab-value normalization, negation/uncertainty, temporal, relation-sparsity, missing-UMLS-relation, overly-broad semantic type, overly-narrow mapping, and answer-choice abstraction mismatch. For each: definition, a MedQA example, effect on symbolic scoring, how an LLM may bridge it, and how the LLM may still fail.

Empirical anchor (verified seed). Canonical concepts appear under many surface forms in this benchmark — dyspnea: 6 forms across 113 questions (shortness of breath 57, dyspnea 28, difficulty breathing 21, ...); chest pain: 6 forms across 55 questions; myocardial infarction: 5 forms (incl. STEMI 10, “heart attack”). Exact matching to a single canonical surface would systematically miss equivalent mentions, the mechanistic root of symbolic brittleness. [TBD-EXP8: distribution of mismatch types over symbolic failures and the hard-core set, with κ .]

9. Discussion

The thesis is that **UMLS grounding alone is insufficient and LLM reasoning alone is not trustworthy enough**; the research value lies in how a hybrid system uses structured knowledge to **audit, constrain, and explain** while relying on the LLM to bridge semantic gaps. We expect symbolic-only methods to underperform, UMLS-in-prompt to give limited gains (context $\neq$ constraint), verification to help weak proposers, and naive revision to risk harm — so the productive role of symbolic knowledge is as a **verifier and error detector** (§7.5) and an **explanation substrate**, not a standalone solver. An **error taxonomy** for the hybrid system (LLM-right/symbolic-wrong and vice versa, verifier true/false positives, grounding/relation failures, revision harm) will indicate which failures future agentic systems must address. We will state clearly where gains are small, mixed, or category-specific.

10. Limitations

MedQA is exam-style, not clinical practice; multiple choice may overstate reasoning. UMLS is incomplete and not a reasoning engine; concept-extraction and entity-linking errors propagate downstream; symbolic relations encode causal/temporal reasoning poorly. LLM rationales may be unfaithful, so explanation metrics need manual validation. Hybrid gains may be category-dependent, and answer revision can introduce harm. The neuro-symbolic results here are **pending experiments** (the system is not yet run in this snapshot), and the preliminary numbers quoted from the repository README are unverified. The system provides no medical advice or clinical decision support.

11. Future Work

This paper sets up **Paper 3’s agentic neuro-symbolic diagnosis**: from static answer selection toward **adaptive evidence acquisition** that chooses which clinical evidence to seek next; **category-specific reasoning modules**; **probabilistic belief updates** over a differential; **contradiction-aware planning** using the verifiers developed here; **differential diagnosis** rather than single-answer selection; **medical knowledge-graph expansion** to reduce concept mismatch; and **executable, interpretable evidence traces**. The verification and grounding components built for Paper 2 become the trustworthiness layer of that agent.

12. Conclusion

We frame and study the concept mismatch problem and the broader question of how structured biomedical knowledge can ground and verify LLM medical reasoning. Using a category-balanced MedQA benchmark, we compare LLM-only, symbolic-only, and hybrid systems and characterize where symbolic grounding and verification help versus fail. The expected and seed-supported finding is that symbolic knowledge is most valuable not as a solver but as an **auditor** of LLM reasoning — detecting contradictions, weak grounding, and likely errors — while the LLM supplies the semantic flexibility symbolic matching lacks. Data, configurations, and analysis code are released; we make no clinical-utility claims.

Appendix / Reproducibility (outline)

- A. UMLS index manifest (release, concept/relation counts) — EXP1.
- B. Grounding and prompt templates — EXP2/EXP4.
- C. Verifier definitions and thresholds — EXP6.
- D. Full per-method, per-category tables — EXP3-EXP7, EXP10.
- E. Concept-mismatch annotation guide and κ — EXP8.
- F. Error-prediction features and model — EXP9.

Figures and tables (asset map)

- Table 1 dataset distribution · Table 2 system variants · Table 3 overall accuracy by method · Table 4 category accuracy by method · Table 5 grounding/verification metrics · Table 6 concept-mismatch taxonomy · Table 7 ablations.
- Figure 1 pipeline (figures/fig1_pipeline.png, ready) · Figure 2 grounding example [TBD-EXP2] · Figure 3 accuracy LLM-only/symbolic/hybrid [TBD-EXP3/7] · Figure 4 category-gain heatmap [TBD-EXP7] · Figure 5 grounding coverage vs correctness [TBD-EXP9] · Figure 6 violation score vs error probability [TBD-EXP9] · Figure 7 concept-mismatch examples [TBD-EXP8]. Seed figure: groundable surface vs difficulty (figures/fig_seed_phrase_vs_acc.png).

Publication strategy (notes)

arXiv-first. Strong fits: biomedical-informatics venues (AMIA), ML-for-health (ML4H, CHIL), clinical-NLP, and neuro-symbolic / knowledge-representation workshops. Minimum to submit: EXP1-EXP3 + EXP6 + EXP9 (symbolic-insufficient, verification-helps, scores-predict-error). EXP4/5/7 complete the hybrid comparison; EXP8/10 add the concept-mismatch depth and mechanism attribution reviewers will want.